Table 1. Data sources and their roles

Dataset / endpoint	Cases	Controls	Role
eQTLGen blood cis-eQTL	—	—	Exposure; up to 31,684 participants (median 28,092 per variant); predominantly European
BBJ GD (GCST90018627)	2,809	172,656	Discovery; East Asian
FinnGen GD (E4_GRAVES_STRICT)	3,962	496,386	Post hoc same-disease comparison; Finnish
UKB hyperthyroidism (GCST90038636)	3,731	480,867	Broader-phenotype comparison; self-reported hyperthyroidism or thyrotoxicosis; ancestry not established as exclusively European
FinnGen GO (GRAVES_OPHT)	858	499,490	Original ophthalmopathy (TED-enriched) comparison; Finnish
FinnGen GO strict (E4_GRAVES_OPHT_STRICT)	753	499,595	Post hoc ophthalmopathy definition sensitivity; Finnish
GTEx v8 tissue eQTL	—	—	Post hoc tissue extension (thyroid, cultured fibroblasts, subcutaneous and visceral adipose, whole blood; n = 469–670; Table S4)
1000 Genomes Phase 3 European reference (n = 503)	—	—	LD clumping of instruments and reference allele frequencies

The three FinnGen R12 endpoints share participants and are not independent replications. FinnGen ophthalmopathy controls are all other participants, including people with GD but no recorded eye disease, so GO estimates do not estimate TED risk among patients with GD. UKB hyperthyroidism is broader than GD. ‘Strict’ is the FinnGen endpoint label, not proof of perfect specificity. The descriptive orbital RNA-sequencing series (four TED patients, one control) is separate from these datasets. eQTL, expression quantitative trait locus; LD, linkage disequilibrium; TED, thyroid eye disease; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

Table 2. MR and colocalization results for *TSHR*, *IGF1R* and *CTLA4* across five disease datasets

Gene	Dataset	IVs	OR (95% CI)	*P* value	Q-test *P*	PP.H2	PP.H3	PP.H4	PP.H4, p12 = 10^{-6}
TSHR	BBJ GD	1	0.12 (0.07–0.21)	1.09×10^{-14}*†	NA	<0.001	0.049	**0.951**	0.661
TSHR	FinnGen GD	1	0.08 (0.05–0.12)	3.19×10^{-32}*†	NA	<0.001	0.736	0.264	0.035
TSHR	UKB hyperthyroidism	1	0.09 (0.06–0.14)	8.77×10^{-28}*†	NA	<0.001	0.774	0.226	0.028
TSHR	FinnGen GO	1	0.10 (0.04–0.24)	2.82×10^{-7}*†	NA	<0.001	0.014	**0.986**	**0.875**
TSHR	FinnGen GO strict	1	0.15 (0.06–0.39)	9.02×10^{-5}*	NA	0.027	0.082	**0.891**	0.451
IGF1R	BBJ GD	4	1.56 (1.07–2.28)	0.0212*	0.902	0.690	0.236	0.073	0.008
IGF1R	FinnGen GD	3	1.07 (0.86–1.33)	0.539	0.882	0.731	0.259	0.010	0.001
IGF1R	UKB hyperthyroidism	4	1.35 (1.07–1.70)	0.0117*	0.250	0.400	0.196	0.404	0.063
IGF1R	FinnGen GO	3	1.41 (0.85–2.33)	0.182	0.309	0.623	0.346	0.032	0.003
IGF1R	FinnGen GO strict	3	1.52 (0.92–2.49)	0.0996	0.409	0.571	0.391	0.038	0.004
CTLA4	BBJ GD	1	0.18 (0.11–0.27)	5.45×10^{-15}*†	NA	<0.001	0.799	0.201	0.025
CTLA4	FinnGen GD	2	0.17 (0.08–0.36)	2.15×10^{-6}*†	0.0275	<0.001	0.746	0.254	0.033
CTLA4	UKB hyperthyroidism	2	0.21 (0.08–0.57)	0.00220*	0.00131	<0.001	0.047	**0.953**	0.672
CTLA4	FinnGen GO	2	0.17 (0.04–0.68)	0.0118*	0.0486	<0.001	0.022	**0.978**	**0.816**
CTLA4	FinnGen GO strict	2	0.17 (0.05–0.64)	0.00888*	0.0755	<0.001	0.021	**0.979**	**0.820**

MR estimates are ORs per approximately one SD higher genetically predicted blood expression: Wald ratio for one IV, multiplicative random-effects IVW for several. *TSHR* used one instrument, rs179252, in every dataset; its G allele raises blood expression by 0.105 SD per allele (F = 176), and its per-allele disease ORs were 0.77–0.82 (Table S3). The per-SD ORs therefore rescale a moderate per-allele association, and each single-IV *P* value equals the variant's disease-association z-test *P* value. *CTLA4* multi-IV estimates are concentrated in rs13030124 and heterogeneous (Q-test *P*); in BBJ and UKB, two of four single-instrument omissions removed nominal *IGF1R* significance (Figure S2). * Nominal $P < 0.05$; † $P < 0.05/2{,}544$, the screen threshold, applied descriptively outside BBJ. PP.H2–H4 are colocalization posterior probabilities (H2, expression signal only; H3, distinct variants; H4, one shared variant) with $p1 = p2 = 10^{-4}$ and $p12 = 10^{-5}$ unless stated; bold, ≥0.80. They are not *P* values. BBJ colocalization pairs European eQTL with East Asian disease statistics. FinnGen ophthalmopathy controls are all other participants, including people with GD but no recorded eye disease, so GO estimates do not estimate TED risk among patients with GD. CI, confidence interval; IV, instrumental variable; IVW, inverse-variance weighted; MR, Mendelian randomization; NA, not applicable (one IV); OR, odds ratio; PP, posterior probability; SD, standard deviation; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.

Table 3. Thirteen BBJ discoveries and their FinnGen GD comparison

BBJ discovery gene	Chr	IVs, BBJ/FinnGen GD	BBJ OR (95% CI)	BBJ *P*	FinnGen GD OR (95% CI)	FinnGen GD *P*	Focused support	FinnGen GD PP.H4
HLA-A	6 (MHC)	4/4	1.98 (1.73–2.27)	2.60×10^{-23}	1.08 (0.77–1.53)	0.648	No	<0.001
HLA-DQA2	6 (MHC)	1/1	2.39 (1.95–2.93)	8.30×10^{-17}	1.90 (1.64–2.20)	1.41×10^{-17}	Yes	<0.001
CTLA4	2	1/2	0.18 (0.11–0.27)	5.45×10^{-15}	0.17 (0.08–0.36)	2.15×10^{-6}	Yes	0.254
TSHR	14	1/1	0.12 (0.07–0.21)	1.09×10^{-14}	0.08 (0.05–0.12)	3.19×10^{-32}	Yes	0.264
C4A	6 (MHC)	1/3	0.45 (0.35–0.57)	2.23×10^{-10}	0.41 (0.26–0.63)	4.97×10^{-5}	Yes	<0.001
HSD3B7	16 (16p11.2 cluster)	1/2	0.29 (0.18–0.47)	2.04×10^{-7}	0.90 (0.72–1.13)	0.372	No	0.011
TUBB	6 (MHC)	2/3	1.49 (1.28–1.74)	3.14×10^{-7}	3.28 (2.09–5.17)	2.68×10^{-7}	Yes	<0.001
VKORC1	16 (16p11.2 cluster)	1/0	0.13 (0.06–0.29)	6.35×10^{-7}	NA	NA	NA	0.011
TNFSF14	19	1/1	0.63 (0.52–0.76)	1.48×10^{-6}	0.77 (0.65–0.92)	0.00361	Yes	0.303
PRSS36	16 (16p11.2 cluster)	1/1	6.00 (2.87–12.57)	2.02×10^{-6}	1.40 (1.00–1.96)	0.0490	No	0.063
MAPKAPK5	12	4/4	0.977 (0.967–0.987)	5.28×10^{-6}	0.990 (0.929–1.055)	0.751	No	0.041
PSMB8	6 (MHC)	1/1	1.25 (1.13–1.38)	6.86×10^{-6}	0.96 (0.85–1.08)	0.512	No	<0.001
IFNGR1	6	1/1	2.10 (1.51–2.91)	9.41×10^{-6}	1.62 (1.18–2.24)	0.00330	Yes	0.313

Genes passed the BBJ screen threshold ($P < 0.05/2{,}544$) and are listed by BBJ *P* value. Focused support in FinnGen GD requires the same direction and $P < 0.05/13$; 7 of 13 qualify. IVs are counted separately for each dataset because instrument availability differs. The five MHC genes and the three chromosome 16p11.2 genes are not independent signals; the latter are compatible with one regional association. *MAPKAPK5* is shown to three decimals to preserve its small effect. NA, not estimable (no retained instrument), not a negative result; *VKORC1* colocalization does not require an MR instrument. PP.H4, shared-variant posterior probability under $p12 = 10^{-5}$; bold, ≥0.80. Chr, chromosome of the strongest instrument; MHC, major histocompatibility complex; CI, confidence interval; IV, instrumental variable; OR, odds ratio; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.